Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 20: Perfect Matchings

Bijections, Inverses, and a Loop Closed

Chartrand & Zhang, Section 5.4

Name: _____ Date: _____

Where we are

One-to-one says nothing is hit twice. Onto says nothing is missed. A function with both properties pairs up the two sets perfectly — and exactly those functions can be run backwards.

By the end of today you should be able to

1. Prove a function is bijective by proving both properties.
2. State when an inverse function exists, and compute one.
3. Describe the equivalence relation a function induces on its domain.

1. Bijections

Definition

$f : A \to B$ is **bijective** (a **bijection**) if it is both one-to-one and onto.
A bijection from A to A is called a **permutation** of A.

A proof that f is bijective has two parts — one for each property. Label them.

Worked (Result 5.59). $f : \mathbb{R}^+ \to \mathbb{R}^+$, $f(x) = \sqrt{x}$.

One-to-one. Assume $f(a) = f(b)$, so $\sqrt{a} = \sqrt{b}$. Squaring both sides gives $a = b$.

Onto. Let $r \in \mathbb{R}^+$. Then $r^2 \in \mathbb{R}^+$ and $f(r^2) = \sqrt{r^2} = r$.

Therefore f is bijective. ■

How the onto half was found

The proof produces r^2 out of nowhere. It was not guessed: solve $\sqrt{x} = r$ for x, getting $x = r^2$, *then* write the proof forwards.
Scratch work runs backwards; proofs are written forwards. That gap is why finished proofs often look like magic, and knowing it is there is half of learning to write them.

2. Inverses

Theorem

A function $f : A \to B$ has an inverse $f^{-1} : B \to A$ **if and only if** f is bijective. In that case f^{-1} is also bijective, and

$$f^{-1}\big(f(a)\big) = a \quad \text{for all } a \in A, \qquad f\big(f^{-1}(b)\big) = b \quad \text{for all } b \in B.$$

Why both properties are needed. To define $f^{-1}(b)$ we must send b back to the element that maps to it. That requires:

Requirement	Supplied by
At least one such element exists	Onto — otherwise $f^{-1}(b)$ is undefined.
At most one such element exists	One-to-one — otherwise the output is not unique.

Those are exactly the two conditions for f^{-1} to be a function, from Monday. **Bijective is not an extra hypothesis — it is precisely what is needed.**

Compute one. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 3x + 7$.

Solve $y = 3x + 7$ for x: $x = (y - 7)/3$.

So $f^{-1}(y) = (y - 7)/3$. Check: $f(f^{-1}(5)) = f(-2/3) = 5$.

3. Compositions of bijections

Theorem

Let $f : A \to B$ and $g : B \to C$.

- If f and g are one-to-one, so is $g \circ f$.
- If f and g are onto, so is $g \circ f$.
- If f and g are bijective, so is $g \circ f$.

Sketch the first. Assume $(g \circ f)(a) = (g \circ f)(b)$, so $g(f(a)) = g(f(b))$. Since g is one-to-one, $f(a) = f(b)$. Since f is one-to-one, $a = b$. ■

Two applications of the working form, peeled from the outside in.

4. Closing a loop: the relation a function induces

Definition

For any function $f : A \to B$, define a relation on A by

$$a \sim b \quad \text{if and only if} \quad f(a) = f(b).$$

This is always an equivalence relation. Fill in why:

Property	Reason
Reflexive	$f(a) = f(a)$, since equality is reflexive.
Symmetric	If $f(a) = f(b)$ then $f(b) = f(a)$.
Transitive	If $f(a) = f(b)$ and $f(b) = f(c)$ then $f(a) = f(c)$.

All three are inherited from equality.

Last week's observation, now a theorem

In Practice Set 16 you noticed that "same birthday," "same file extension," and "same truth column" are all equivalence relations. Now you can see why: each has the form $f(a) = f(b)$ for some function f.

The equivalence classes are the preimages. The class of a is the set of everything mapping where a maps.

And the connection to today: f is one-to-one exactly when every equivalence class is a single element, because no two distinct inputs share an image.

Try it. $f : \{1, 2, 3, 4\} \to \{a, b\}$ with $f = \{(1, a), (2, b), (3, a), (4, a)\}$.

Classes: $\{1, 3, 4\}$ and $\{2\}$ One-to-one? No

Exit ticket

1. Why does a function need to be bijective to have an inverse? One sentence per property.

2. Find f^{-1} for $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 4x - 1$.

Before Monday

Read Section 5.5 and review the Chapter 5 Highlights, pp. 191–193. The Key Results page collects everything from the last two weeks.

Self-check (not collected). Section 5.4, Exercises 5, 11, and 15 — odd-numbered, answers in the back.

Coming up

Homework 7 is due tonight, Friday, October 2, 11:59 p.m.
Quiz 7 is Monday, October 5 — 15 minutes on Sections 5.3 and 5.4.
Homework 8 covers Sections 5.3 and 5.4 and is due Friday, October 9.

Questions I still have